

Nicole Robiro

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RESEARCH FIELDS

Primary: Development Economics, Agricultural Economics

Secondary: Applied Econometrics, Environmental Economics, Health Economics

Methods and data: Impact evaluation, causal inference, remote sensing and geospatial data

Regional focus: Sub-Saharan Africa, South Asia

EDUCATION

Goethe University Frankfurt

Frankfurt, Germany

Ph.D. in Economics

2020 – expected June 2027

- Dissertation: *Remote Sensing Applications in Agriculture and Health Economics*
- Supervisors: Prof. Matthias Schündeln, Ph.D.

Goethe University Frankfurt

Frankfurt, Germany

M.Sc. in International Economics and Economic Policy

2017 – 2019

- Thesis: The Impact of Development Projects on Forest Loss in Africa: Evidence from Remote Sensing Data (with Prof. Matthias Schündeln, Ph.D.)

Universidade Católica Portuguesa

Lisbon, Portugal

Academic stay abroad

2016

Goethe University Frankfurt

Frankfurt, Germany

B.Sc. in Economics and Business Administration

2013 – 2017

- Exchange semester, Universidade Católica Portuguesa, Lisbon, Portugal (2016)

ACADEMIC POSITIONS

Goethe University Frankfurt, Chair of Development Economics

Frankfurt, Germany

Research and Teaching Associate

Sep 2021 – present

- Research on agricultural productivity, health and social unrest in Sub-Saharan Africa and South Asia using satellite and other novel geospatial data.
- Teaching and supervision at Ph.D., master's and bachelor's level (see Teaching).
- Organisation of research workshops and Ph.D. conferences (see Service).

PUBLICATIONS IN REFEREED JOURNALS

“Predictable Patterns of Protest: The Impact of Agricultural Cycles on Social Unrest in Kenya”
(with S. Seelos). *Food Policy*, 2026, doi.org/10.1016/j.foodpol.2026.103136.

WORKING PAPERS

“The Productivity Effects of Irrigation Aid: Evidence from Border Difference-in-Differences in Sub-Saharan Africa.” **Job Market Paper**. August, 2026

Abstract: This paper examines the microlevel effectiveness of irrigation-focused development aid in sub-Saharan Africa between 2005 and 2020 using georeferenced project data and high-resolution remote sensing. Employing a localized difference-in-differences strategy near country borders, the study finds evidence that irrigation aid raises the production value of cash crops, while staple crop yields remain largely unresponsive. This pattern indicates that the gains are concentrated in market-oriented production rather than extending broadly across crop categories. The study contributes to the literature on foreign aid and agricultural development by introducing a high-resolution empirical framework that leverages border matching and grid-level data to identify local treatment effects. This approach provides an alternative identification strategy to commonly used instrumental-variable and shift-share designs, expanding the scope for evaluating geographically targeted development interventions.

“Can Financing Protected Areas Save Forest Cover and Biodiversity? Results from a Geospatial Matching Approach” (with Y. Eilers, J. Kluve, J. Schielein, M. Wong). January, 2026

WORK IN PROGRESS

“Prenatal Exposure and the Pandemic: Assessing the Effects of COVID-19-Driven Air Quality Improvements in India” (with N. Galinski).

POLICY REPORTS AND OTHER PUBLICATIONS

Reports and policy briefs for KfW Development Bank, Financial Cooperation Evaluation Department.

- Policy Brief: Impact Evaluation - Water Sector: Initial Results from the Simiyu Climate Resilience Project in Tanzania (with S. Lange, L. Lenz and S. Prediger), October 2025
- Policy Brief: Willingness to Pay for Water Supply Project in Tanzania (with S. Lange, L. Lenz and S. Prediger), October 2025
- 2-Pager: Beware of Doppelgänger! Synthetic Control Methods in Causal Impact Evaluation (with S. Lange), June 2025
- Pre-Study Report: Impact Evaluation of the Rural Water Supply Component of the Simiyu Climate Resilience Project in Tanzania (with J. Kluve, L. Lenz and S. Prediger), January, 2025

CONFERENCE PRESENTATIONS AND INVITED TALKS

German Development Economics Conference (GDEC/EDEG), Kiel Institute for the World Economy	2026
Ph.D. Conference in Economics, University of Warwick	2026
Frankfurt-Heidelberg-Mannheim (FHM+) Workshop in Development Economics, University of Stuttgart	2025
GDE Ph.D. Workshop, Kiel Institute for the World Economy	2025
Doctorissimes, Paris School of Economics	2025
GODAD Workshop on Global Development, University of Göttingen	2025

FHM+ Workshop in Development Economics, University of Mannheim	2024
German Development Economics Conference (GDEC), University of Hannover	2024
FHM+ Workshop in Development Economics, Goethe University Frankfurt	2023
Bonn-Frankfurt-Mannheim Ph.D. Conference, Goethe University Frankfurt	2023
Frankfurt-Mannheim Ph.D. Conference, Goethe University Frankfurt	2022
FHM+ Workshop in Development Economics, University of Heidelberg	2022

TEACHING EXPERIENCE

Goethe University Frankfurt	Frankfurt, Germany
– <i>Teaching Assistant</i> , Development Microeconomics (Ph.D. level)	2022 – 2025
– <i>Co-Instructor</i> , Seminar Research Methods (B.Sc level)	2023

Thesis Supervision	2021 – present
– Supervised 14 bachelor's theses and 4 master's theses	

PROFESSIONAL EXPERIENCE

KfW Development Bank, Financial Cooperation Evaluation Department Frankfurt, Germany
Consultant Sep 2023 – Dec 2025

- Managed impact evaluations in the water sector in Tanzania, including the design and coordination of on-site fieldwork.
- Conducted geospatial matching analysis of the effects of KfW-financed protected areas on forest cover and biodiversity in Latin America.
- Drafted policy briefs and analytical reports for senior management and government partners.

GIZ (Deutsche Gesellschaft für Internationale Zusammenarbeit) Eschborn, Germany
Student Assistant, Competence Centre Digital Society – Data4Development Feb 2021 – Jul 2021

- Built an internal evidence database in Excel and Power BI and acted as technical contact for its rollout and extension.
- Reviewed evidence-based methods and workflows across GIZ operations.

GIZ (Deutsche Gesellschaft für Internationale Zusammenarbeit) Eschborn, Germany
Intern, Competence Centre Digital Society – Data4Development Jul 2020 – Jan 2021

- Contributed to the evaluation of an e-registry project for health care centres in Malawi using Stata.
- Advised on an activity monitoring tool for the special initiative “A World Without Hunger”.
- Investigated remote sensing data for the estimation of regional crop yields.

Deutsche Bundesbank, Financial Stability Department Frankfurt, Germany
Intern Jan 2020 – Apr 2020

- Analysed risk indicators for the non-bank financial sector in Matlab and Stata and automated recurring estimation routines.

KfW Development Bank, Financial Cooperation Evaluation Department Frankfurt, Germany
Student Assistant Jan 2018 – Dec 2019

- Synthesised evaluation evidence across health and education projects to assess overall effectiveness.

- Analysed blended finance, project performance against the fragility index, and project effects using night-time lights data.

FIELDWORK

Water sector impact evaluation, Tanzania 2024

- Supported field implementation of the household survey: supervised enumerator teams on site, ran spot checks and back-checks on incoming interviews, and verified questionnaire items.

PROFESSIONAL SERVICE AND ACTIVITIES

Mentee, Mentoring Hessen Pro.Career 2023

Conference Organisation

Lead Organiser, Frankfurt-Mannheim-Heidelberg Workshop 2023

Co-Organiser, Bonn-Frankfurt-Mannheim Ph.D. Conference 2023

Lead Organiser, Frankfurt-Mannheim Ph.D. Conference 2022

Mentor, Shaping Pathways 2022

Erasmus Student Network (ESN), Frankfurt

Vice President and Local Representative 2016 – 2017

Treasurer 2014 – 2015

SKILLS AND CERTIFICATIONS

Software: Stata, R, Python, Matlab, Google Earth Engine, QGIS, L^AT_EX, Power BI, VBA, MS Office

Languages: German (native), English (C1), Polish (C1), Spanish (B2), French (A2), Portuguese (A2), Kiswahili (A2)

REFERENCES

Prof. Matthias Schündeln, Ph.D
Chair of Development Economics
Goethe University Frankfurt
schuendeln@wiwi.uni-frankfurt.de

Prof. Dr. Reyn van Ewijk
Chair of Statistics and Econometrics
Johannes Gutenberg Universität Mainz
vanewijk@uni-mainz.de

Dr. Simon Lange
KfW Development Bank
Frankfurt, Germany
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